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213. On the Development of the Disturbing Function in the Lunar Theory *(continued)*

Tables for the lunar part of the disturbing function (continued)

[PDF p. 326; printed p. 304.] We have

$$\Omega_2 = m'\frac{a^4}{a'^5}\left(\tfrac{3}{8} - \tfrac{33}{8}\eta^2 + \tfrac{75}{8}\eta^4\right) \text{ multiplied into}$$

Lunar part *infra.*		
$\frac{73}{6}e''^6$	cos	$-4g'\}$
$\frac{55}{8}e''^5$		$-3g'$
$3e''^4 + \frac{11}{4}e''^6$		$-2g'$
$1 + 2e''^2$		$-g'$
$* e'' + \frac{8}{3}e''^3$		$0g' \quad -\mathfrak{C}'$
$1 + \frac{5}{2}e''^2$		$+g'$
$\frac{11}{8}e''^2$		$+2g'$
$\frac{55}{12}e''^3$		$+3g'\}$
* Exact value is $e'(1-e'^2)^{-\frac{3}{2}}$.		
$\frac{7}{128}e^8$	Solar part *suprà.*	$-3g$
$\frac{1}{6}e^6$		$-2g$
$\frac{11}{8}e^5 + \frac{7}{48}e^7$		$-g$
$-\frac{5}{2}e - \frac{15}{8}e^3$ (exact)		$0g$
$1 + 2e^2 + \frac{61}{64}e^4$		$+g \quad +\mathfrak{C}$
$-\frac{1}{2}e + e^3$		$+2g$
$-\frac{3}{8}e^2 + \frac{11}{16}e^4$		$+3g$
$-\frac{7}{24}e^3$		$+4g$
$-\frac{95}{384}e^4$		$+5g\}$

$g - g' + \mathfrak{C} - \mathfrak{C}'$ parallactic equation.

[PDF p. 327; printed p. 305.] We have

$$\Omega_3 = m'\frac{a^4}{a'^5}\left(\tfrac{5}{8} - \tfrac{15}{8}\eta^2 + \tfrac{15}{8}\eta^4\right) \text{ multiplied into}$$

Lunar part *infra.*		
$\frac{165}{4}e''^6$	cos	$-6g'\}$
$\frac{177}{8}e''^5$		$-5g'$
$5e''^4 + \frac{22}{1}e''^6$		$-4g'$
$1 + 6e''^2$		$-3g' \quad -3\mathfrak{C}'$
$-e'' + \frac{8}{3}e''^3$		$-2g'$
$\frac{1}{8}e''^2$		$-g'$
0 (exact)		$0g'\}$
$\frac{75}{128}e^8$	Solar part *suprà.*	$-g$
$-\frac{35}{8}e^7$ (exact)		$0g$
$\frac{57}{8}e^6 - \frac{65}{16}e^8$		$+g$
$-\frac{9}{2}e^5 + \frac{35}{4}e^7$		$+2g$
$1 - 6e^2 + \frac{593}{64}e^4$		$+3g \quad +3\mathfrak{C}$
$\frac{3}{2}e^3 + \frac{57}{8}e^5$		$+4g$
$\frac{15}{8}e^4 - \frac{135}{16}e^6$		$+5g$
$\frac{9}{4}e^5$		$+6g$
$\frac{343}{128}e^6$		$+7g\}$

[PDF p. 328; printed p. 306.] We have

$$\Omega_4 = m'\frac{a^4}{a'^5}\left(\tfrac{9}{4}\eta^2 - \tfrac{15}{2}\eta^4\right) \text{ multiplied into}$$

Lunar part *infra.*		
$\frac{23}{12}e''^2$	cos	$-2g'\}$
$\frac{11}{8}e''^3$		$-g'$
$*\, e'' + \frac{8}{3}e''^3$		$0g' \quad -\mathfrak{C}'$
$1 + 2e''^2$		$+g'$
$3e''^2 + \frac{11}{4}e''^4$		$+2g'$
$\frac{55}{8}e''^3$		$+3g'$
$\frac{77}{6}e''^4$		$+4g'\}$
* Exact value is $e'(1-e'^2)^{-\frac{3}{2}}$.		
$\frac{7}{128}e^8$	Solar part *suprà.*	$-3g$
$\frac{5}{6}e^6$		$-2g$
$\frac{11}{8}e^5 + \frac{7}{48}e^7$		$-g$
$-\frac{5}{2}e - \frac{15}{8}e^3$ (exact)		$0g$
$1 + 2e^2 + \frac{61}{64}e^4$		$+g \quad +\mathfrak{C}$
$-\frac{1}{2}e + e^3$		$+2g$
$-\frac{3}{8}e^2 + \frac{11}{16}e^4$		$+3g$
$-\frac{7}{24}e^3$		$+4g$
$-\frac{95}{384}e^4$		$+5g\}$

[PDF p. 329; printed p. 307.] We have

$$\Omega_5 = m' \frac{a^4}{a'^5} \left(\tfrac{15}{8}\eta^2 - \tfrac{15}{4}\eta^4 \right) \text{ multiplied into}$$

Lunar part *infra.*		
$\frac{77}{6}e''^4$	cos	$-4g'\}$
$\frac{55}{8}e''^3$		$-3g'$
$3e''^2 + \frac{11}{4}e''^4$		$-2g'$
$1 + 2e''^2$		$-g'$
$* \, e'' + \frac{8}{3}e''^3$		$0g' \quad -\mathfrak{C}'$
$\frac{11}{8}e''^3$		$+g'$
$\frac{23}{12}e''^2$		$+2g'\}$
* Exact value is $e'(1-e'^2)^{-\frac{3}{2}}$.		
$\frac{75}{128}e^8$	Solar part *suprà.*	$-g$
$-\frac{35}{8}e^7$ (exact)		$0g$
$\frac{57}{8}e^6 - \frac{65}{16}e^8$		$+g$
$-\frac{9}{2}e^5 + \frac{35}{4}e^7$		$+2g$
$1 - 6e^2 + \frac{593}{64}e^4$		$+3g \quad +3\mathfrak{C}$
$\frac{3}{2}e^3 + \frac{57}{8}e^5$		$+4g$
$\frac{15}{8}e^4 - \frac{135}{16}e^6$		$+5g$
$\frac{9}{4}e^5$		$+6g$
$\frac{343}{128}e^6$		$+7g\}$

[PDF p. 330; printed p. 308.] We have

$$\Omega_6 = m' \frac{a^4}{a'^5}\left(\tfrac{15}{8}\eta^2 - \tfrac{15}{4}\eta^4\right) \text{ multiplied into}$$

Lunar part *infra.*		
$\frac{165}{4}e''^6$	cos	$-6g'\}$
$\frac{177}{8}e''^5$		$-5g'$
$5e''^4 + 22e''^6$		$-4g'$
$1 + 6e''^2$		$-3g' \quad -3\mathfrak{C}'$
$-e'' + \frac{8}{3}e''^3$		$-2g'$
$\frac{1}{8}e''^2$		$-g'$
0 (exact)		$0g'\}$
$\frac{7}{128}e^8$	Solar part *suprà.*	$-3g$
$\frac{1}{6}e^6$		$-2g$
$\frac{11}{8}e^5 + \frac{7}{48}e^7$		$-g$
$-\frac{5}{2}e - \frac{15}{8}e^3$ (exact)		$0g$
$1 + 2e^2 + \frac{61}{64}e^4$		$+g \quad +\mathfrak{C}$
$-\frac{1}{2}e + e^3$		$+2g$
$-\frac{3}{8}e^2 + \frac{11}{16}e^4$		$+3g$
$-\frac{7}{24}e^3$		$+4g$
$-\frac{95}{384}e^4$		$+5g\}$

[PDF p. 331; printed p. 309.] The arrangement of the tables hardly requires any explanation; each line of the upper half of a table is to be read in combination with each line of the lower half of the same table. Thus a term of Ω_1 is

$$m'\frac{a^4}{a'^5}(\tfrac{3}{4}\eta^2+\tfrac{9}{4}\eta^4)(-\tfrac{47}{4}e''^4)(1-\tfrac{3}{2}e''^2+\tfrac{1}{16}e''^4)\cos(-g-2g'+2\mathfrak{C}-2\mathfrak{C}').$$

It is to be observed that in the table for Ω_1 (where the arguments depend only on the mean anomalies), the several terms (other than the constant term) occur in pairs of equal terms, having respectively the arguments $ig+i'g'$ and $-ig-i'g'$; such equal terms are to be united together, or, what is the same thing, the coefficients must be multiplied by 2. Thus we have the two terms

$$m'\frac{a^4}{a'^5}(\tfrac{3}{4}\eta^2+\tfrac{9}{4}\eta^4)(-\tfrac{47}{4}e''^4)(1-\tfrac{3}{2}e''^2+\tfrac{1}{16}e''^4)2\cos(-6g+5g'),$$

$$\text{Ditto}\qquad\qquad \cos(\ \ 6g+5g'),$$

or, what is the same thing, the term is

$$m'\frac{a^4}{a'^5}(\tfrac{3}{4}\eta^2+\tfrac{9}{4}\eta^4)(-\tfrac{47}{4}e''^4)(1-\tfrac{3}{2}e''^2+\tfrac{1}{16}e''^4)2\cos(\ \ 6g+5g').$$

This is the case even when either i or i' vanishes; but, as already noticed, it is not the case for the constant term where i and i' both vanish. There are not any such equal pairs in the tables for Ω_2, &c., where all the arguments contain a part independent of the mean anomalies. The quantity $\mathfrak{C}$ occurs in the upper or solar half of the table; but it is to be recollected that $\mathfrak{C}\,(=\varpi-\theta)$ involves θ, which is an element of the moon's orbit.

The peculiar form in which the coefficients are exhibited, viz. as the product of a term depending on the inclination, a term depending on the eccentricity of the moon's orbit, and a term depending on the eccentricity of the sun's orbit, is not to be considered as a want of completion of the development; it is, on the contrary, an important advantage.

[The Errata in the Tables, noticed *Mem. R. Ast. Soc.* vol. XXVIII. p. 216, have been corrected. The values of P_1', Q_1', Q_2', &c. used in the construction of the Tables *ante* p. 298 may also be obtained from the Tables of the Developments of Functions in *The Theory of Elliptic Motion*, p. 216.]

Addition.

I deduce from the preceding an expression for the disturbing function in a form similar to and easily comparable with the forms given by Sir J. W. Lubbock in his work *On the Theory of the Moon* &c. (London, 1834), pp. 30–35, and by Pontécoulant in the *Théorie Analytique du Système du Monde*, t. IV. (Paris, 1846), pp. 58–61. The several notations are as follows, viz.:

[PDF p. 332; printed p. 310.]

Lubbock.	Pontécoulant.	Supra.
ξ,	ϕ,	g
ξ',	ϕ',	g'
τ,	ξ,	$g - g' + \mathfrak{C} - \mathfrak{C}'$
η,	η,	$g + \mathfrak{C}$
γ,	γ,	$2\eta\sqrt{1-\eta^2} \div (1-2\eta^2)$

(γ, the tangent of the inclination, is employed by the above-named two authors instead of η, the sine of the semi-inclination, the relation between these two quantities gives to γ^2 or η^4

$$\gamma^2 = 4\eta^2 + 12\eta^4,$$

and conversely,

$$\eta^2 = \tfrac{1}{4}\gamma^2 - \tfrac{1}{16}\gamma^4,$$

$$\eta^4 = \tfrac{1}{16}\gamma^4,$$

which are useful for replacing one of these quantities by the other). The disturbing function is taken by Lubbock with the contrary sign, and he includes in the before-mentioned omitted term depending only on the sun's radius vector, that is, R (Lubbock) $= -m'\dfrac{1}{r'} - \Omega$; he uses also, in reference to the sun, subscript strokes instead of accents (so that, in referring to his notation, $\xi_|$ should properly be here used in the place of ξ', but this difference is obviously immaterial). Pontécoulant's disturbing function is taken with the same sign as in the present memoir, or we have R (Pontécoulant) $= \Omega$. Lubbock and Pontécoulant give in the principal terms the part involving $m'\dfrac{a^4}{a'^5}$, which depends on the square of the parallax, and which was disregarded in the preceding development. I have in the sequel inserted these terms from Lubbock's expression. We have, in fact:

[PDF p. 333; printed p. 311.]

$$\Omega = m' \frac{a^2}{a'^3} \text{ multiplied into}$$

	Lubbock.		Pontécoulant.		Arguments ut *suprà.*
	Arguments.	Nos.	Arguments.	Nos.	
$\frac{1}{4} + \frac{3}{4}e^2 + \frac{9}{64}e^4 + \frac{9}{2}e'^2 + \frac{15}{32}e'^4 + \cdots$		0		0	
$-\frac{3}{2}e - \frac{27}{16}e^3 - \frac{15}{64}e^5 + \cdots$	τ	1	2ξ	30	$2g - 2g' + 2\mathfrak{C} - 2\mathfrak{C}'$
$-\frac{1}{2}e + \frac{1}{16}e^3 - \frac{5}{4}ee'^2 + \cdots$	ξ	2	ϕ	1	g
$\frac{15}{8}e^2 - \frac{45}{32}e^4 - \frac{15}{4}ee' + \cdots$	$\tau - \xi$	3	$2\xi - \phi$	31	$g - 2g'$
$-\frac{3}{4}ee' - \frac{45}{16}e^3e' + \cdots$	$\tau + \xi$	4	$2\xi + \phi$	32	$3g - 2g'$
$\frac{3}{4}e^2 - \frac{15}{32}e^4 \cdots$	ξ'	5	ϕ'	6	
$\frac{3}{8}e^2 - \frac{105}{16}e^4 + \cdots$	$\tau - \xi'$	7	$2\xi - \phi'$	33	$2g - 3g'$
$\frac{15}{8}ee' + \cdots$	$\tau + \xi'$	8	$2\xi + \phi'$	34	$2g - g'$
$-\frac{3}{4}ee' \cdots$	$2\xi'$	9	$2\phi'$	2	$2g'$
$\frac{15}{16}e^2 - \frac{15}{4}ee'^2 \cdots$	$\tau - 2\xi'$	9	$2\xi - 2\phi'$	35	$-2g'$
$\frac{1}{2}ee' \cdots$	$\tau + 2\xi'$	10	$2\xi + 2\phi'$	36	$4g - 2g'$
$\frac{3}{16}e^3 + \frac{9}{4}ee'^2 \cdots$	$\xi - \xi'$	11	$\phi + \phi'$	9	$g - g'$

(Table continues in original; arguments to No. 41 with full coefficients.)

* Variation. † Evection.

[PDF p. 334; printed p. 312.]

$$\Omega\,(\text{continued}) = m'\frac{a^2}{a'^3} \text{ multiplied into}$$

	Lubbock.		Pontécoulant.		Arguments
	Arguments.	Nos.	Arguments.	Nos.	ut *suprà.*
$-\frac{1}{16}e^3$	3ξ	20	3ϕ	3	$3g$
$-\frac{3}{32}e^2$	$\tau-3\xi$	21	$2\xi-3\phi$	43	$-g-2g'$
$\frac{15}{32}e^2$	$\tau+3\xi$	22	$2\xi+3\phi$	44	$5g-2g'$
$\frac{3}{4}ee'$	$2\xi+\xi'$	23	$2\phi+\phi'$	11	$2g+g'$
$\frac{105}{16}e^2e'$	$\tau-2\xi+\xi'$	24	$2\xi-2\phi+\phi'$	45	g'
$-\frac{3}{8}ee'$	$2\xi-\xi'$	26	$2\phi-\phi'$	12	$2g-g'$
$\frac{3}{4}e^2e'$	$\tau-2\xi-\xi'$	28	$2\xi-2\phi-\phi'$	46	$-g'$
$-\frac{15}{16}ee'^2$	$\tau+2\xi-\xi'$	28	$2\xi+2\phi-\phi'$	47	$4g-g'$
$\frac{9}{16}e^3$	$\xi+2\xi'$	29	$\phi+2\phi'$	13	$g+2g'$
$\frac{45}{32}e^2e'$	$\tau-\xi-2\xi'$	30	$2\xi-\phi-2\phi'$	49	$g-4g'$
$-\frac{9}{16}ee'^2$	$\xi-2\xi'$	32	$\phi-2\phi'$	14	$g-2g'$
$\frac{3}{32}e^3$	$\tau-\xi+2\xi'$	34	$2\xi-\phi+2\phi'$	51	$3g-4g'$

(Remainder of table continues numbering through No. 53.)

[PDF p. 335; printed p. 313.]

$$\Omega\,(\text{continued}) = m'\frac{a^2}{a'^3} \text{ multiplied into}$$

	Lubbock.		Pontécoulant.		Arguments
	Arguments.	Nos.	Arguments.	Nos.	ut *suprà.*
$-\frac{15}{16}e^2e'$	$\tau+3\xi-\xi'$	43			$5g-3g'+2\mathfrak{C}-2\mathfrak{C}'$
$\frac{3}{32}e^3$	$3\xi-\xi'$	44	$3\phi-\phi'$	16	$3g-g'$
$\frac{3}{32}e^3$	$\tau-3\xi-\xi'$	45	$2\xi-3\phi+\phi'$	54	$-g-3g'+2\mathfrak{C}-2\mathfrak{C}'$
$\frac{175}{64}e^2$	$\tau+3\xi-\xi'$	46			$5g-3g'$
$\frac{15}{16}e^4$	$2\xi+2\xi'$	47	$2\phi+2\phi'$	15	$2g+2g'$
$-\frac{255}{16}e^4e'$	$\tau-2\xi+2\xi'$	48	$2\xi-2\phi+2\phi'$	55	$4g'$
$-\frac{3}{8}e^4e'$	$2\xi-2\xi'$	49	$2\phi-2\phi'$	14	$2g-2g'$
$\frac{45}{8}e^4e'$	$\tau+2\xi-2\xi'$	52	$2\xi+2\phi-2\phi'$	56	$4g-2g'$
$\frac{535}{64}e^4e'$	$\tau-2\xi-2\xi'$	53			$-3g'+2\mathfrak{C}-2\mathfrak{C}'$
$\frac{8}{64}ee'^3$	$\xi'-3\xi'$	54			$-5g'$
$\frac{45}{8}e^4e'$	$\xi+3\xi'$	56			$g-5g'$
$\frac{77}{64}e^4$		59			$g-3g'$
$-\frac{545}{64}e^4e'$	$\tau-4\xi'$	61			$2g-6g'$
$-\frac{1}{32}ee'$	$\tau+4\xi'$	62			$2g+2g'$

(Numbering continues to the last argument of the Lunar half.)

[PDF p. 336; printed p. 314.]

$$\Omega\,(\text{continued}) = m'\frac{a^2}{a'^3}\text{ multiplied into}$$

	Lubbock.		Pontécoulant.		Arguments
	Arguments.	Nos.	Arguments.	Nos.	ut *suprà*.
$-\frac{3}{4}\eta^2 ee'$	$\tau-\xi-2\eta$ rev.	67	$2\xi-\phi-2\eta$ rev.	59	$g+2g'$
$-\frac{3}{4}\eta^2 ee'$	$\tau+\xi-2\eta$ rev.	69	$2\xi+\phi-2\eta$ rev.	61	$-g+2g'$
$\frac{9}{4}\eta^2 e'$	$\xi'-2\eta$ rev.	71	$\phi'-2\eta$ rev.	21	$2g-g'$
$\frac{9}{4}\eta^2 e'$	$\xi'+2\eta$ rev.	72	$\phi'+2\eta$ rev.	22	$2g+g'$
$-\frac{3}{4}\eta^2 e^2 e'$	$\tau-\xi'-2\eta$ rev.	73	$2\xi-\phi'-2\eta$ rev.	63	$3g'$
$\frac{3}{4}\eta^2 e^2 e'$	$\tau+\xi'-2\eta$ rev.	75	$2\xi+\phi'-2\eta$ rev.	65	g'
$\frac{15}{4}\eta^2$	$2\xi-2\eta$	77	$2\phi-2\eta$	24	$0g$
$\frac{15}{4}\eta^2$	$2\xi+2\eta$	78	$2\phi+2\eta$	24	$4g$
$\frac{3}{4}\eta^2 ee'$	$\tau-\xi-2\eta$ rev.	79	$2\xi-2\phi-2\eta$ rev.	67	$2g+2g'$
$-\frac{3}{4}\eta^2 ee'$	$\tau+\xi-2\eta$ rev.	81	$2\xi+2\phi-2\eta$ rev.	69	$2g-2g'$
$\frac{27}{4}\eta^2 ee'$	$\xi+\xi'-2\eta$	83	$\phi-\phi'-2\eta$ rev.	77	$g-g'$
$\frac{9}{4}\eta^2 ee'$	$\xi+\xi'+2\eta$	84	$\phi+\phi'+2\eta$	28	$3g+g'$

(Numbering reaches 97 at end of this page.)

[PDF p. 337; printed p. 315.]

$$\Omega\,(\text{continued}) = m'\frac{a^2}{a'^3}\ \text{multiplied into}$$

*(Lunar half of Ω ends with No. 100, table preceded by * Parallactic equation.)*					
	Lubbock.		Pontécoulant.		Arguments
	Arguments.	Nos.	Arguments.	Nos.	ut *suprà.*
$*\frac{3}{4}+\frac{3}{4}e^2+\frac{3}{4}e'^2-\frac{33}{8}\eta^2$	τ	101	ξ	70	$g-g'$
$-\frac{15}{16}e$	$\tau-\xi$	102	$\xi-\phi$	71	$-g'$
$-\frac{3}{16}e$	$\tau+\xi$	103	$\xi+\phi$	72	$g-g'$
$\frac{6}{8}e'$	$\tau-\xi'$	104	$\xi-\phi'$	73	$g-2g'$
$\frac{6}{8}e'$	$\tau+\xi'$	105	$\xi+\phi'$	74	g
$-\frac{135}{64}ee'$	$\tau-2\xi'$	106	$\xi-2\phi$	75	$-g-g'$
$\frac{9}{64}ee'$	$\tau+2\xi'$	108	$\xi-\phi-\phi'$	76	$3g-g' \quad \mathfrak{C}-\mathfrak{C}'$
$-\frac{45}{16}ee'^2$	$\tau-\xi'+\xi'$	109	$\xi+\phi-\phi'$	77	$2g$
$-\frac{3}{16}ee'^2$	$\tau+\xi'-\xi'$	110	$\xi+\phi+\phi'$	79	$g+g'$
$\frac{9}{16}ee'^2$	$\tau-\xi'-\xi'$	111	$\xi-\phi+\phi'$	78	$g+0g$
$\frac{135}{64}ee'^2$	$\tau+\xi'+\xi'$	112	$\xi-2\phi+\phi'$		$g+g'$
$\frac{33}{64}\eta^2$	$\tau-2\eta$	113			g
$\frac{9}{4}\eta^2$	$\tau+2\eta$	114			g

(Solar half continues to No. 130 in the original.)

[PDF p. 338; printed p. 316.]

$$\Omega\,(\text{continued}) = m'\frac{a^2}{a'^3}\text{ multiplied into}$$

	Lubbock.		Pontécoulant.		Arguments
	Arguments.	Nos.	Arguments.	Nos.	ut *suprà.*
$\frac{75}{64}e^2$	$3\tau+3\xi$	122			$5g-3g'$
$-\frac{225}{16}e^2$	$3\tau-\xi-\xi'$	123			$2g-4g'$
$-\frac{15}{16}ee'^2$	$3\tau+\xi+\xi'$	124			$4g-2g'$
$\frac{45}{16}e^2e'$	$3\tau-\xi+\xi'$	125			$2g-2g' \quad +2\mathfrak{C}-2\mathfrak{C}'$
$\frac{75}{16}e^2e'$	$3\tau-\xi-\xi'$	126			$4g-4g'$
$\frac{635}{64}\eta^2$	$3\tau-2\xi$	127			$3g-5g'$
$\frac{3}{64}\eta^2$	$3\tau+2\xi$	128			$3g-g'$
$\frac{15}{8}\eta^2$	$3\tau-2\eta$	129			$g-5g' \quad +\mathfrak{C}-3\mathfrak{C}'$

where the abbreviation *rev.* denotes that the argument to which it is attached has its sign reversed in the third column of arguments: thus Arg. 63 (Lubbock), it is $-(2\tau-2\eta)$ which is equal to $2\eta+2\mathfrak{C}$. As the formula contains only cosines, there is, of course, no change in the sign of the coefficient.

On comparing with Lubbock's value, I find some differences, which are as follows:–It will be recollected that R (Lubbock) $= -m'\frac{1}{r'} - \Omega$, so that the signs of the coefficients of Ω are to be reversed in order to deduce the corresponding coefficients of R. The Nos. refer to Lubbock's arguments, and the exterior factor $m'\frac{a^2}{a'^3}$ or $m'\frac{a^4}{a'^5}$ is disregarded[1].

Arg. 1. Lubbock's coefficient (viz. the coefficient in R) is

$$-\tfrac{3}{2}\Big(1-\tfrac{5}{8}e^2-\tfrac{5}{8}e''^4+\tfrac{15}{4}e^2+\tfrac{3}{4}e'^2+\tfrac{1}{2}\tfrac{e^4}{e'^4}+\tfrac{1}{12}\tfrac{a^2}{a'^2}\Big)\cos^4\tfrac{1}{2}i.$$

[1] I have been favoured with a note from Sir J. W. Lubbock, confirming my values of the coefficients for the arguments 8, 18, 58, 101 and 123.—Added 12th Feb. 1859, A. C.

[PDF p. 339; printed p. 317.] which, substituting for $\cos^4 \frac{1}{2}i$ its value $1 - \frac{1}{2}\gamma^2 + \frac{7}{16}\gamma^4 - \&c.$, and developing to the fourth order, gives

$$-\tfrac{3}{2}\Big(1 - \tfrac{5}{8}e^2 - \tfrac{5}{8}e''^4 + \tfrac{15}{4}e^2e''^2 + \tfrac{15}{16}e^4e''^4 - \tfrac{1}{2}\gamma^2 + \tfrac{1}{2}\gamma^2e^2 + \tfrac{1}{2}\gamma^2e''^2 + \tfrac{7}{16}\gamma^4 + \tfrac{1}{12}\tfrac{a^2}{a'^2}\Big)e^2,$$

which, in fact, agrees with the value given *suprà*. I have only referred to this term in order to make the reduction.

Arg. 8, Lubbock's coefficient is

$$-\tfrac{1}{8}\big(1 - \tfrac{5}{8}e^2 + \tfrac{3}{4}e''^2 - \tfrac{3}{4}\gamma^2\big)e^2;$$

the exterior sign should be + instead of −. The term is given with the correct sign, Pontécoulant, Arg. 2.

Arg. 18, Lubbock's coefficient is

$$-\tfrac{15}{4}\big(1 - \tfrac{3}{2}e^2 - \tfrac{105}{4}e^4e''^2 - \tfrac{1}{2}\gamma^2\big)\cos^4 \tfrac{1}{2}i,$$

which, developed to γ^4, would be

$$-\tfrac{15}{4}\big(1 - \tfrac{3}{2}e^2 - \tfrac{105}{4}e^4e''^2 - \gamma^2\big).$$

I make it

$$-\tfrac{15}{4}\big(1 - \tfrac{3}{2}e^2 - \tfrac{105}{4}e^4e''^2 - \tfrac{1}{2}\gamma^2\big).$$

The remaining differences are

No. of Arg. Lubbock.	Lubbock's coefficient.	Coefficient from Development *suprà.*
58	$+\frac{45}{64}e^4e'^2$	$-\frac{845}{64}e^4e'^2$
59	$+\frac{591}{64}e^4$	$-\frac{77}{32}e^4$
60	$-\frac{7453}{128}e^4$	$-\frac{27}{32}e^4e'^2$
61	$+\frac{741}{128}e^4$	$-\frac{1}{32}e^4e'^2$
∗62	$-\frac{3}{8}\big(1-\frac{5}{2}e^2+\frac{3}{2}e^4-\gamma^2\big)\gamma^2$	$-\frac{3}{8}\big(1-\frac{5}{2}e^2+\frac{3}{2}e^4-\gamma^2\big)\gamma^2$

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No. of Arg. Lubbock.	Lubbock's coefficient.	Coefficient from Development *suprà*.
*63	$-\frac{3}{8}\left(1+\frac{3}{2}e^2-\frac{5}{2}e''^2+\frac{5}{2}\gamma^2\right)\gamma^2$	$-\frac{3}{8}\left(1+\frac{3}{2}e^2-\frac{5}{2}e''^2-\gamma^2\right)\gamma^2$
95	$-\frac{27}{16}\gamma^2 e''^3$	$-\frac{27}{32}\gamma^2 e''^3$
96	$-\frac{27}{16}\gamma^2 e''^3$	$-\frac{27}{32}\gamma^2 e''^3$
*101	$+\frac{3}{8}\left(1+3e''^2+3e^2-\frac{11}{4}\gamma^2\right)\gamma^2$	$-\frac{3}{8}\left(1+2e''^2+2e^2-\frac{11}{4}\gamma^2\right)\gamma^2$
115	$-\frac{15}{128}\gamma^2$	$-\frac{15}{32}\gamma^2$
123	$-\frac{225}{16}ee'$	$+\frac{225}{16}ee'$

The greater part of the discordant terms do not occur in Pontécoulant's development, which is not carried so far, and the only differences which I find in the coefficients of Pontécoulant $R\,(=\Omega)$ are as regards the arguments 18, 57, 70, corresponding respectively to Lubbock's arguments 62, 63, 101, included in the preceding table, and for which Pontécoulant's coefficients, correcting for the change of sign, correspond with those given by Lubbock. But I see no room for a mistake in the preceding investigation as regards the coefficients of these three terms; the names in η of the coefficients of 62 and 63 are simply the quantity $(\frac{1}{2}\eta^2-\frac{1}{8}\eta^4)$, which forms the exterior factor of Ω_3 and Ω_6 respectively, and which, putting for η^2 and η^4 their values, is equal to $\frac{1}{8}(1-\gamma^2)\gamma^2$, and as regards the coefficient of 101, the question being to obtain by the mere multiplication of the factors $1+2e^2+2e''^2$ &c., and $1+2e''^2$ set opposite to $g+\mathfrak{C}$ and $0g'-\mathfrak{C}'$ respectively in the table for Ω_2.

2, *Stone Buildings, W.C., 7th July*, 1858.

[PDF p. 341; printed p. 319.]

214. The First Part of a Memoir on the Development of the Disturbing Function in the Lunar and Planetary Theories

[From the *Memoirs of the Royal Astronomical Society*, vol. XXVIII. (1860), pp. 187–215. Read November 10, 1858.]

THE development, as is well known, depends upon that of the reciprocal of the distance of the two planets; and Hansen's Memoir "Entwickelung der negativen und ungeraden Potenzen der Quadratwurzel der Function $r^2 + r'^2 - 2rr'(\cos U \cos U' + \sin U \sin U' \cos J)$," *Abh. der K. Sächs. Ges. zu Leipzig*, t. II., pp. 286–376 (1854), contains a formula which is truly fundamental, viz. the expression of the coefficient of the general term

$$\frac{r^n}{r'^{n+1}} \cos(jU + j'U')$$

of the development of the reciprocal of the distance as expressed in the above-mentioned form, where r, r' are the radius vectors of the inferior and superior planets respectively, and U, U' are the angular distances from the mutual node. In the lunar theory, where the higher powers of $\frac{r}{r'}$ are neglected, we have in this manner a small number of terms each of which is to be separately developed in multiple cosines of the mean anomalies. This can be effected as in the *Fundamenta Nova*, and my "Memoir on the Development of the Disturbing Function in the Lunar Theory," *R. Ast. Soc. Mem.*, t. XXVII., 1859, [213], a mere completion of which is Hansen's process [1]. In fact if f, f' are the true anomalies, and $\mathfrak{C}$, $\mathfrak{C}'$ the distances

[1] I take the opportunity of mentioning the memoir of Hansen's which immediately precedes that above referred to, viz. "Entwickelung des Products einer Potenz des Radius Vectors mit dem Sinus oder Cosinus eines Vielfachen der wahren Anomalie in Reihen die nach den Sinussen oder Cosinussen der Vielfachen der wahren der excentrischen oder mittleren Anomalie fortschreiten," t. II., pp. 183–281 (1853).

[PDF p. 342; printed p. 320.] of the pericentres from the mutual node, then we have $U = f + \mathfrak{C}$, $U' = f' + \mathfrak{C}'$, and the general term is

$$\frac{r^n}{r'^{n+1}} \cos(jf + j'f' + j\mathfrak{C} + j'\mathfrak{C}')$$

where r, f are given functions of the mean anomaly g, and r', f' are the like functions of the mean anomaly g'. And the development depends upon those of

$$r^n {}^{\cos}_{\sin} jf, \qquad r'^{-n-1} {}^{\cos}_{\sin} jf',$$

which (if we consider as well negative as positive values of the index n) are each of the form

$$r^n {}^{\cos}_{\sin} jf,$$

and when the developments of these expressions are known, we obtain at once by the mere addition and subtraction of the coefficients of the cosines and sines of the different multiples of g and g', the development of

$$\frac{r^n}{r'^{n+1}} \cos(jf + j'f' + j\mathfrak{C} + j'\mathfrak{C}')$$

in the tabular form employed in my memoir just referred to. In the planetary theory, we must unite together the terms containing the different powers of $\frac{r}{r'}$ so as to form the entire coefficient $D(j, j')$ of $\cos(jU + j'U')$; if then we write

$$r = a(1 + x), \qquad r' = a'(1 + x'),$$

and develope the coefficient in powers of x, x', we have the general term

$$x^\alpha x'^{\alpha'} \cos(jU + j'U')$$

which admits of development in multiple cosines of the mean anomalies, in the same manner precisely as the before-mentioned general term

$$\frac{r^n}{r'^{n+1}} \cos(jU + j'U')$$

in the lunar theory. It is proper to remark that this method is really identical with that commonly made use of in the planetary theory: the only difference is, that by Hansen's fundamental formula, we have the complete expression of the coefficient $D(j, j')$ developed in powers of sin or of tan $\frac{1}{2}\phi$, instead of (as in the ordinary methods) the first two or three terms of this development.

[PDF p. 343; printed p. 321.] The required development of

$$x^{\alpha} x'^{\alpha'} \cos(jU + j'U')$$

or, what is the same thing,

$$x^{\alpha} x'^{\alpha'} \cos(jf + j'f' + j\mathfrak{C} + j'\mathfrak{C}')$$

depends on the developments of

$$x^{\alpha}{}^{\cos}_{\sin} jf, \qquad x'^{\alpha'}{}^{\cos}_{\sin} jf'.$$

These are functions of the same form, and we may consider only

$$x^{\alpha}{}^{\cos}_{\sin} jf.$$

The value of x is $\left(\frac{r}{a} - 1\right)$ and we could of course calculate

$$\left(\frac{r}{a} - 1\right)^{\alpha}{}^{\cos}_{\sin} jf$$

by the methods of the *Fundamenta Nova* or the memoir of Hansen's referred to in the foot-note. But if we write $f = g + y$ (y is the equation of the centre), then the required expressions depend on

$$x^{\alpha}{}^{\cos}_{\sin} jy$$

which are actually calculated as far as e^7 for $\alpha = 0, 1, 2, \ldots, 7$, and j an undetermined symbol, by Leverrier in the *Annales de l'Observatoire de Paris*, t. I. pp. 346–348 (1855). Hence, by the mere substitution, in Leverrier's formula, of the numerical values of j and j', and by the addition and subtraction of the coefficients of the cosines or sines of the different multiples of g and g', we may obtain the development of

$$x^{\alpha} x'^{\alpha'} \cos(jf + j'f' + j\mathfrak{C} + j'\mathfrak{C}')$$

in a tabular form similar to that employed in my memoir already referred to.

I have thought it desirable to put together the various results above referred to, and to investigate by a different process the expression for Hansen's coefficient, which in his memoir is obtained by means of a long series of transformations which it is not very easy to follow, and is not exhibited in quite the most simple form. And this is what I have done in the present first part of a memoir on the development of the disturbing function. My object has been to exhibit, in as complete a form as possible, the preliminary development in multiple cosines of the true anomalies; and to indicate the process of the ulterior development in multiple cosines of the mean anomalies.

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I.

Let $\mathfrak{M}$ be the inferior, $\mathfrak{M}'$ the superior of the two planets (in the lunar theory $\mathfrak{M}$ is the moon, $\mathfrak{M}'$ the sun) and let the quantities relating to the two bodies be for $\mathfrak{M}$,

r, the radius-vector,
U, the distance from node,
$\mathfrak{C}$, the distance of pericentre from node,
f, the true anomaly,
g, the mean anomaly,
a, the mean distance,
e, the eccentricity,

and for $\mathfrak{M}'$ the accented letters, r', &c., in the like significations.

The node referred to is the ascending node of the orbit of $\mathfrak{M}$ upon that of $\mathfrak{M}'$, and I write also

Φ, the inclination of the orbit of $\mathfrak{M}$ to that of $\mathfrak{M}'$,
η, $= \sin \frac{1}{2}\Phi$.

The disturbing function is in the first instance given as a function of r, r', H, where

$$\cos H = \cos U \cos U' + \sin U \sin U' \cos \Phi,$$

that is, as a function of r, r', U, U', Φ, or, what is the same thing, of r, r', U, U', η. And the preliminary development is a development in multiple cosines of U, U'. We have then

$$U = f + \mathfrak{C},$$
$$U' = f' + \mathfrak{C}',$$

and finally

$$r = a \text{ elqr}(e, g),$$
$$f = \text{elta}(e, g),$$
$$r' = a' \text{ elqr}(e', g'),$$
$$f' = \text{elta}(e', g'),$$

and the ulterior development is a development in multiple cosines of g, g', $\mathfrak{C}$, $\mathfrak{C}'$, the coefficients involving, as before, η, and also a, e, a', e'. But as usual it is not attempted to carry the development further, by introducing in the coefficients in place of the relative quantities $\mathfrak{C}$, $\mathfrak{C}'$, η, the remaining elements of the two orbits, which, if it were necessary to use them, would be

for $\mathfrak{M}$,

θ, the longitude of node,
σ, the departure of node,
ϕ, the inclination,
ϖ, the departure of pericentre,

and for $\mathfrak{M}'$ the like accented quantities, the orbit of reference being any fixed or moveable orbit whatever.

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II.

The expression for the disturbing function on $\mathfrak{M}$ (that is, when the superior planet disturbs the inferior) is

$$\mathfrak{M}' \left\{ -\frac{r\cos H}{r'^2} + \frac{1}{\sqrt{r^2+r'^2-2rr'\cos H}} \right\}$$

and that for the disturbing function on $\mathfrak{M}'$ (that is, when the inferior planet disturbs the superior) is

$$\mathfrak{M} \left\{ -\frac{r'\cos H}{r^2} + \frac{1}{\sqrt{r^2+r'^2-2rr'\cos H}} \right\}$$

where the disturbing function is taken with Lagrange's sign ($= -R$, if R be the disturbing function of the *Mécanique Céleste*).

But we may in the first instance consider the development of the reciprocal of the distance of the two planets

$$= \frac{1}{\sqrt{r^2+r'^2-2rr'\cos H}}.$$

The preceding expression for $\cos H$ may be written

$$\cos H = \cos U \cos U' + \sin U \sin U' \, (1-2\eta^2),$$

or in either of the two forms

$$\begin{aligned} \cos H &= \cos(U-U') - 2\eta^2 \sin U \sin U', \\ \cos H &= (1-\eta^2)\cos(U-U') + \eta^2 \cos(U+U'). \end{aligned}$$

Now imagine the function developed in ascending powers of $\dfrac{r^n}{r'^{n+1}}$, the coefficient of $\dfrac{r^n}{r'^{n+1}}$ will contain $\cos^n H$, $\cos^{n-2} H$, $\ldots$ to $\cos H$ or 1 according as n is even or odd; and if we then substitute for $\cos H$ the given last expression, and express the different powers of $\cos H$ in multiple cosines of $U-U'$ and $U+U'$, and make the final expression contain the cosines of opposite arguments each with the same coefficient, it is easy to see that the form of the general term is

$$\frac{r^n}{r'^{n+1}} C_n(j,j') \cos(jU+j'U'),$$

where j, j' each of them extend through the values n, $n-2$, $\ldots -n$, and where

$$C_n(-j,-j') = C_n(j,j').$$

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[PDF p. 346; printed p. 324.] Thus in particular for $n = 0, 1, 2, 3$, the combinations (j, j') belonging to the several arguments are,

$0,0$	$1,1 \quad -1,1$ $1,-1 \quad -1,-1$	$2,2 \quad 0,2$ $2,0 \quad 0,0$ $2,-2 \quad 0,-2$	$3,3 \quad 1,3$ $3,1 \quad 1,1$ $3,-1 \quad 1,-1$ $3,-3 \quad 1,-3$

But as the coefficients C satisfy the condition $C(-j, -j') = C(j, j')$, the two terms $C(j, j') \cos(jU + j'U')$ and $C(-j, -j') \cos(-jU - j'U')$ are equal to each other, and they may be combined together into a single term. The general term may consequently be written

$$\frac{r^n}{r'^{n+1}} 2C_n(j, j') \cos(jU + j'U'),$$

where j has only the values $n, n-2, \ldots 1$ or 0, and j' has as before the values $n, n-2, \ldots, -n$; except (which occurs only when n is even) for $j = 0$, when j' has only the values $n, n-2, \ldots 0$: and in the particular case $j = j' = 0$, the last-mentioned expression for the general term must be multiplied by $\frac{1}{2}$, or, what is the same thing, the factor 2 must be omitted. In particular for $n = 0, 1, 2, 3, 4$, the combinations (j, j') belonging to the several arguments are

$0,0$	$1,1$ $1,-1$	$2,2 \quad 0,2$ $2,0 \quad 0,0$ $2,-2$	$3,3 \quad 1,3$ $3,1 \quad 1,1$ $3,-1 \quad 1,-1$ $3,-3 \quad 1,-3$	$4,4 \quad 2,4$ $4,2 \quad 2,2$ $4,0 \quad 2,0$ $4,-2 \quad 2,-2$ $4,-4 \quad 2,-4$

I remark that in a series $K(j, j')^{\cos}_{\sin}(jU + j'U')$, where each argument occurs positively and negatively, and $K(-j, -j') = \pm K(j, j')$ according as the series is one of cosines

[PDF p. 347; printed p. 325.] or of sines, we may say that the *discrete* general term is $K(j,j')^{\cos}_{\sin}(jU+j'U')$, or that $K(j,j')$ is the *discrete* coefficient of the cosine or sine, but if we unite together the terms with opposite arguments so as to form the general term $2K(j,j')^{\cos}_{\sin}(jU+j'U')$, then this may be called the *concrete* general term, and $2K(j,j')$ may be said to be the concrete coefficient of the cosine or sine. In a sine series the term corresponding to the argument zero vanishes; in a cosine series this is not in general the case, and the concrete term corresponding to the argument zero must be multiplied by $\frac{1}{2}$.

Returning to the question in hand, from the symmetry of the expression to be developed, we have

$$C_n(j,j') = C_n(j',j),$$

and it follows that the only coefficients which need be calculated are those for which j is not negative, and not less in absolute magnitude than j'; the remaining coefficients are respectively equal to coefficients which satisfy these conditions. Thus for $n = 0,1,2,3,4$, the combinations (j,j') corresponding to the coefficients in question are,

$0,0$	$1,1$	$2,2$	$0,0$	$3,3$	$1,1$	$4,4$	$2,2$
	$1,-1$	$2,0$		$3,1$	$1,-1$	$4,2$	$2,0$
		$2,-2$		$3,-1$		$4,0$	$2,-2$
				$3,-3$		$4,-2$	$0,0$
						$4,-4$	$2,-4$

and we have, for instance, $C(2,4) = C(4,2)$, $C(-2,-4) = C(2,4) = C(4,2)$, &c. Under the preceding restriction, viz. j not negative, and not less in absolute magnitude than j', the expression for $C_n(j,j')$ (deduced from the formulæ of Hansen's Memoir) is as follows; viz. putting as usual $\Pi\alpha = 1.2.3\ldots\alpha$, and also $\Pi_1(x-\frac{1}{2}) = \frac{1}{2}\cdot\frac{3}{2}\cdot\frac{5}{2}\ldots(x-\frac{1}{2})$, and representing the hypergeometric series

$$1 + \frac{\alpha\cdot\beta}{1\cdot\gamma}x + \frac{\alpha\cdot\alpha+1\cdot\beta\cdot\beta+1}{1\cdot2\cdot\gamma\cdot\gamma+1}x^2 + \text{\&c.}$$

by $F(\alpha,\beta,\gamma,x)$, we have

$$C(j,j') = \frac{2^{j+j'}\,\Pi_1(\frac{1}{2}(n+j)-\frac{1}{2})\,\Pi_1(\frac{1}{2}(n+j')-\frac{1}{2})}{\Pi\frac{1}{2}(n-j)\,\Pi\frac{1}{2}(n-j')\,\Pi(j+j')}$$

$$\times\eta^{j+j'}(1-\eta^2)^{\frac{1}{2}(j-j')}F(-n+j+1,\; n+j+1,\; j+j'+1,\; \eta^2)$$

[PDF p. 348; printed p. 326.] which, it is to be noticed, is a rational and integral function of η, the highest power being η^n, and the lowest power or order of the coefficient being $\eta^{j+j'}$. I reserve the demonstration of this formula for a separate paragraph.

Let the discrete general term involving $\cos(jU + j'U')$ be represented by

$$D(j, j')\cos(jU + j'U');$$

we have, in like manner as for the coefficients C, $D(-j, -j') = D(j, j')$, $D(j', j) = D(j, j')$, and consequently $D(j, j')$ will be known, if we know its value when the before-mentioned conditions are satisfied; viz. if j be not negative and not less in absolute magnitude than j'; and collecting together the different terms which involve the cosine in question, we find at once, the conditions being satisfied,

$$D(j, j') = \frac{r^j}{r'^{j+1}}\, C_j(j, j') + \frac{r^{j+2}}{r'^{j+3}}\, C_{j+2}(j, j') + \&\text{c.},$$

so that the value of $D(j, j')$ is known. A transformation of this expression will be given in the sequel.

III.

The concrete general term is

$$2D(j, j')\cos(jU + j'U');$$

in particular the concrete terms involving the arguments $U - U'$, $U + U'$, are

$$2D(1, -1)\cos(U - U'), \qquad 2D(1, 1)\cos(U + U'),$$

or, substituting for the D coefficients their values, the terms are

$$2\left\{\frac{r}{r'^2}\, C_1(1, -1) + \frac{r^3}{r'^4}\, C_3(1, -1) + \cdots\right\}\cos(U - U'),$$

$$2\left\{\frac{r}{r'^2}\, C_1(1, 1) + \frac{r^3}{r'^4}\, C_3(1, 1) + \cdots\right\}\cos(U + U').$$

So far I have considered the reciprocal of the radius vector; but if we consider, instead, the disturbing functions, the only difference will be that we must add the term

$$-\frac{r}{r'^2}\cos H,$$

or

$$-\frac{r'}{r^2}\cos H,$$

[PDF p. 349; printed p. 327.] according as the superior planet disturbs the inferior one, or the inferior planet the superior one. The value of $\cos H$ is

$$(1-\eta^2)\cos(U-U') + \eta^2\cos(U+U'),$$

$$= 2C_1(1,-1)\cos(U-U') + 2C_1(1,1)\cos(U+U'),$$

observing that

$$C_1(1,-1) = \tfrac{1}{2}(1-\eta^2), \qquad C_1(1,1) = \tfrac{1}{2}\eta^2;$$

and the terms to be added are consequently, when the superior planet disturbs the inferior,

$$-2\,\frac{r}{r'^2}\,C_1(1,-1)\cos(U-U'),$$

$$-2\,\frac{r}{r'^2}\,C_1(1,1)\cos(U+U'),$$

the effect of which is simply to destroy the same terms contained with the opposite sign in the reciprocal of the distance;

and when the inferior planet disturbs the superior one, the terms to be added are

$$-2\,\frac{r'}{r^2}\,C_1(1,-1)\cos(U-U'),$$

$$-2\,\frac{r'}{r^2}\,C_1(1,1)\cos(U+U'),$$

which are not equal to any terms in the reciprocal of the distance. We may write as follows:

[PDF p. 350; printed p. 328.]

Reciprocal of Distance is,

Disturbing function + Mass of Disturbing Planet is,

	U	$+U'$
$D(0,0)$	cos	0 0
$+2\,D(1,-1)$		1 $-$ 1
(1) $\left[-2\,\frac{r}{r'^2}\,C_1(1,-1)\right]$		1 $-$ 1
(2) $\left[-2\,\frac{r'}{r^2}\,C_1(1,-1)\right]$		1 $-$ 1
$+2\,D(1,1)$		1 1
(1) $\left[-2\,\frac{r}{r'^2}\,C_1(1,1)\right]$		1 1
(2) $\left[-2\,\frac{r'}{r^2}\,C_1(1,1)\right]$		1 1
$+2\,D(2,-2)$		2 $-$ 2
$+2\,D(2,0)$		2 0
$+2\,D(0,2)$		0 2
$+2\,D(2,2)$		2 2
$+2\,D(3,-3)$		3 $-$ 3
$+2\,D(3,-1)$		3 $-$ 1
$+2\,D(1,-3)$		1 $-$ 3
$+2\,D(3,1)$		3 1
$+2\,D(1,3)$		1 3
$+2\,D(3,3)$		3 3
$+2\,D(4,-4)$		4 $-$ 4
$+2\,D(4,-2)$		4 $-$ 2
$+2\,D(2,-4)$		2 $-$ 4
$+2\,D(4,0)$		4 0
$+2\,D(0,4)$		0 4
$+2\,D(4,2)$		4 2
$+2\,D(2,4)$		2 4
$+2\,D(4,4)$		4 4
+&c., &c.		

[1] Only to be inserted for the disturbing function when the superior planet disturbs the inferior, and having the effect of destroying portions of the coefficients of the next preceding terms.
[2] Only to be inserted for the disturbing function when the inferior planet disturbs the superior one.

[End of pp. 326–350 chunk; paper 214 continues in next chunk.]